



DevOps



Kubernetes Dashboard

Kubernetes Dashboard

```
apiVersion: cert-manager.io/v1
kind: ClusterIssuer
metadata:
  name: all-clusterissuer
spec:
  acme:
    server: https://acme-v02.api.letsencrypt.org/directory
    email: leangsengk90@gmail.com
    privateKeySecretRef:
      name: letsencrypt-key
    solvers:
      - http01:
          ingress:
            class: nginx
```

Kubernetes Dashboard

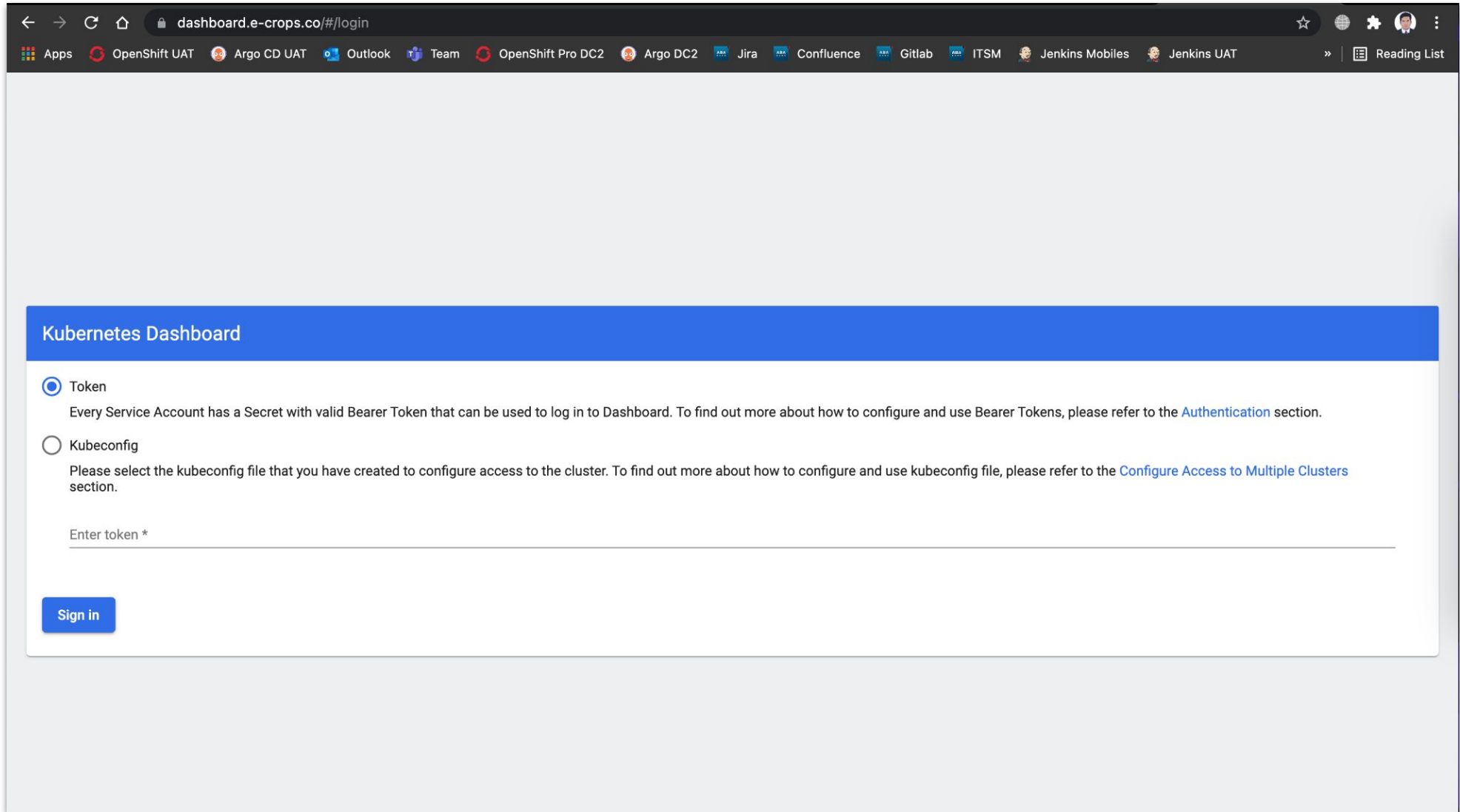
```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: kubernetes-ingress
  namespace: kube-system
  annotations:
    cert-manager.io/cluster-issuer: all-clusterissuer
    kubernetes.io/ingress.class: nginx
    nginx.ingress.kubernetes.io/backend-protocol: "HTTPS"
spec:
  tls:
    - hosts:
        - dashboard.e-crops.co
      secretName: kubernetes-tls
  rules:
    - host: dashboard.e-crops.co
      http:
        paths:
          - pathType: Prefix
            path: "/"
            backend:
              service:
                name: kubernetes-dashboard
                port:
                  number: 443
```

Kubernetes Dashboard

```
apiVersion: v1
kind: Namespace
metadata:
  name: kubernetes-dashboard
---
apiVersion: v1
kind: ServiceAccount
metadata:
  name: admin-user
  namespace: kubernetes-dashboard
---
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  name: admin-user
roleRef:
  apiGroup: rbac.authorization.k8s.io
  kind: ClusterRole
  name: cluster-admin
subjects:
- kind: ServiceAccount
  name: admin-user
  namespace: kubernetes-dashboard
```

Kubernetes Dashboard

- `kubectl -n kubernetes-dashboard create token admin-user --duration=24h`



The screenshot shows a web browser window with the URL `dashboard.e-crops.co/#/login`. The browser's address bar and tabs are visible at the top. The main content area has a blue header bar with the text "Kubernetes Dashboard". Below this, there are two radio button options: "Token" (selected) and "Kubeconfig". The "Token" option has a description: "Every Service Account has a Secret with valid Bearer Token that can be used to log in to Dashboard. To find out more about how to configure and use Bearer Tokens, please refer to the [Authentication](#) section." The "Kubeconfig" option has a description: "Please select the kubeconfig file that you have created to configure access to the cluster. To find out more about how to configure and use kubeconfig file, please refer to the [Configure Access to Multiple Clusters](#) section." Below these options is a text input field labeled "Enter token *" and a blue "Sign in" button.

dashboard.e-crops.co/#/login

Apps OpenShift UAT Argo CD UAT Outlook Team OpenShift Pro DC2 Argo DC2 Jira Confluence Gitlab ITSM Jenkins Mobiles Jenkins UAT Reading List

Kubernetes Dashboard

☒ Token
Every Service Account has a Secret with valid Bearer Token that can be used to log in to Dashboard. To find out more about how to configure and use Bearer Tokens, please refer to the [Authentication](#) section.

☐ Kubeconfig
Please select the kubeconfig file that you have created to configure access to the cluster. To find out more about how to configure and use kubeconfig file, please refer to the [Configure Access to Multiple Clusters](#) section.

Enter token *

Sign in

Create Kubernetes Objects

The screenshot shows the Kubernetes Dashboard interface. The top navigation bar includes the Kubernetes logo, a namespace selector set to 'All namespaces', a search bar, and a red arrow pointing to a '+' icon in the top right corner. The main content area is titled 'Workload Status' and displays five green circles representing different workload types with their respective running pod counts: Daemon Sets (4), Deployments (16), Pods (38), Replica Sets (17), and Stateful Sets (1). A left-hand sidebar lists various Kubernetes resources under categories like Workloads, Service, Config and Storage, and Cluster. At the bottom, a table titled 'Daemon Sets' is partially visible, showing columns for Name, Namespace, Images, Labels, Pods, and Created.

Create Deployment and Service from form

Create from input

Create from file

Create from form

App name *

nginx-apps

10 / 24

Container image *

nginx:1.22.1

Number of pods *

2

Service *

Internal

Port *

80

Target port *

80

Protocol *

TCP

Port

Target port

Protocol *

TCP

Namespace *

default

Deploy

Preview

Cancel

Show advanced options

Create Ingress from input

Create from input Create from file Create from form

Enter YAML or JSON content specifying the resources to create to the currently selected namespace. [Learn more](#)

```
1  apiVersion: networking.k8s.io/v1
2  kind: Ingress
3  metadata:
4    name: nginx-ingress
5  annotations:
6    cert-manager.io/cluster-issuer: all-clusterissuer
7    kubernetes.io/ingress.class: nginx
8  spec:
9    tls:
10     - hosts:
11         - nginx.e-crops.co
12       secretName: nginx-tls
13   rules:
14     - host: nginx.e-crops.co
15       http:
16         paths:
17           - pathType: Prefix
18             path: "/"
19           backend:
20             service:
21               name: nginx-app
22               port:
23                 number: 80
```

Upload Cancel

Modify Kubernetes Objects - Scales

The screenshot shows a Kubernetes dashboard interface with a modal window titled "Scale a resource". The modal is overlaid on a background showing details for a deployment named "nginx-app" in the "default" namespace. The deployment was created on "Jul 5, 2023" and is 39 minutes old. The UID is "4c86fabd-b7e5-458e-9194-100b55c0c3ec". The modal contains the following information:

- Scale a resource**
- Deployment nginx-app will be updated to reflect the desired replicas count.
- Desired replicas ***: 2 (indicated by a blue underline)
- Actual replicas**: 2
- Information message: `kubectl scale -n default deployment nginx-app --replicas=2`
- Buttons: **Scale** and **Cancel**

The background interface also shows sections for "Metadata", "Resource", "Strategy" (RollingUpdate), "Selector" (k8s-app: nginx-app), and "Rolling update strategy" (Max surge: 25%, Max unavailable: 25%).

Modify Kubernetes Objects - Yaml

Metadata

Name
nginx-app

Labels
k8s-app: nginx

Annotations
deployment.

Resource

Strategy
RollingUpd

Selector
k8s-app: nginx

Rolling up

Max surge
25%

Edit a resource

YAML

JSON

```
94     subresource: status
95 spec:
96   replicas: 2
97   selector:
98     matchLabels:
99       k8s-app: nginx-app
100  template:
101    metadata:
102      name: nginx-app
103      creationTimestamp: null
104      labels:
105        k8s-app: nginx-app
106  spec:
107    containers:
108      - name: nginx-app
109        image: nginx:1.22.1
110        resources: {}
111        terminationMessagePath: /dev/termination-log
112        terminationMessagePolicy: File
113        imagePullPolicy: IfNotPresent
114        securityContext:
115          privileged: false
116        restartPolicy: Always
```

i

This action is equivalent to: `kubectl apply -f <spec.yaml>`

Update

Cancel

Deployment / Replica Set / Pod

The screenshot displays the Kubernetes dashboard interface. The top bar shows the 'kubernetes' logo, a 'default' namespace dropdown, a search bar, and user icons. The breadcrumb navigation indicates the path: Workloads > Pods > nginx-app-5cf5ff8858-c6n97. The left sidebar lists various Kubernetes resources, with 'Pods' selected. The main content area features two charts: 'CPU Usage' and 'Memory Usage'. Below these are sections for 'Metadata' and 'Resource information'. Two red arrows point to the 'Pod' icon and the 'Pod' link in the top bar.

Workloads N

- Cron Jobs
- Daemon Sets
- Deployments
- Jobs
- Pods**
- Replica Sets
- Replication Controllers
- Stateful Sets

Service

- Ingresses N
- Ingress Classes
- Services N

Config and Storage

- Config Maps N
- Persistent Volume Claims N
- Secrets N
- Storage Classes

Cluster

- Cluster Role Bindings
- Cluster Roles

CPU Usage

CPU (cores)

0.01
0.005
0

00:09 00:10 00:11 00:12 00:13 00:14 00:15 00:16 00:17 00:18 00:19 00:20 00:21

Memory Usage

Memory (bytes)

2 Mi
0 Mi

00:09 00:10 00:11 00:12 00:13 00:14 00:15 00:16 00:17 00:18 00:19 00:20 00:21

Metadata

Name	Namespace	Created	Age	UID
nginx-app-5cf5ff8858-c6n97	default	Jul 5, 2023	58 minutes ago	bac63d00-a30d-4c57-90d2-2bfcc5400b49

Labels

k8s-app: nginx-app pod-template-hash: 5cf5ff8858

Annotations

cni.projectcalico.org/containerID cni.projectcalico.org/podIP: 10.233.75.37/32 cni.projectcalico.org/podIPs: 10.233.75.37/32

Resource information

Node	Status	IP	QoS Class	Restarts	Service Account
node2	Running	10.233.75.37	BestEffort	0	default

View Pod Log

Workloads > Pods > nginx-app-5cf5ff8858-c6n97 > Logs

Workloads

Cron Jobs

Daemon Sets

Deployments

Jobs

Pods

Replica Sets

Replication Controllers

Stateful Sets

Service

Ingresses

Ingress Classes

Services

Config and Storage

Config Maps

Persistent Volume Claims

Secrets

Storage Classes

Cluster

Cluster Role Bindings

Cluster Roles

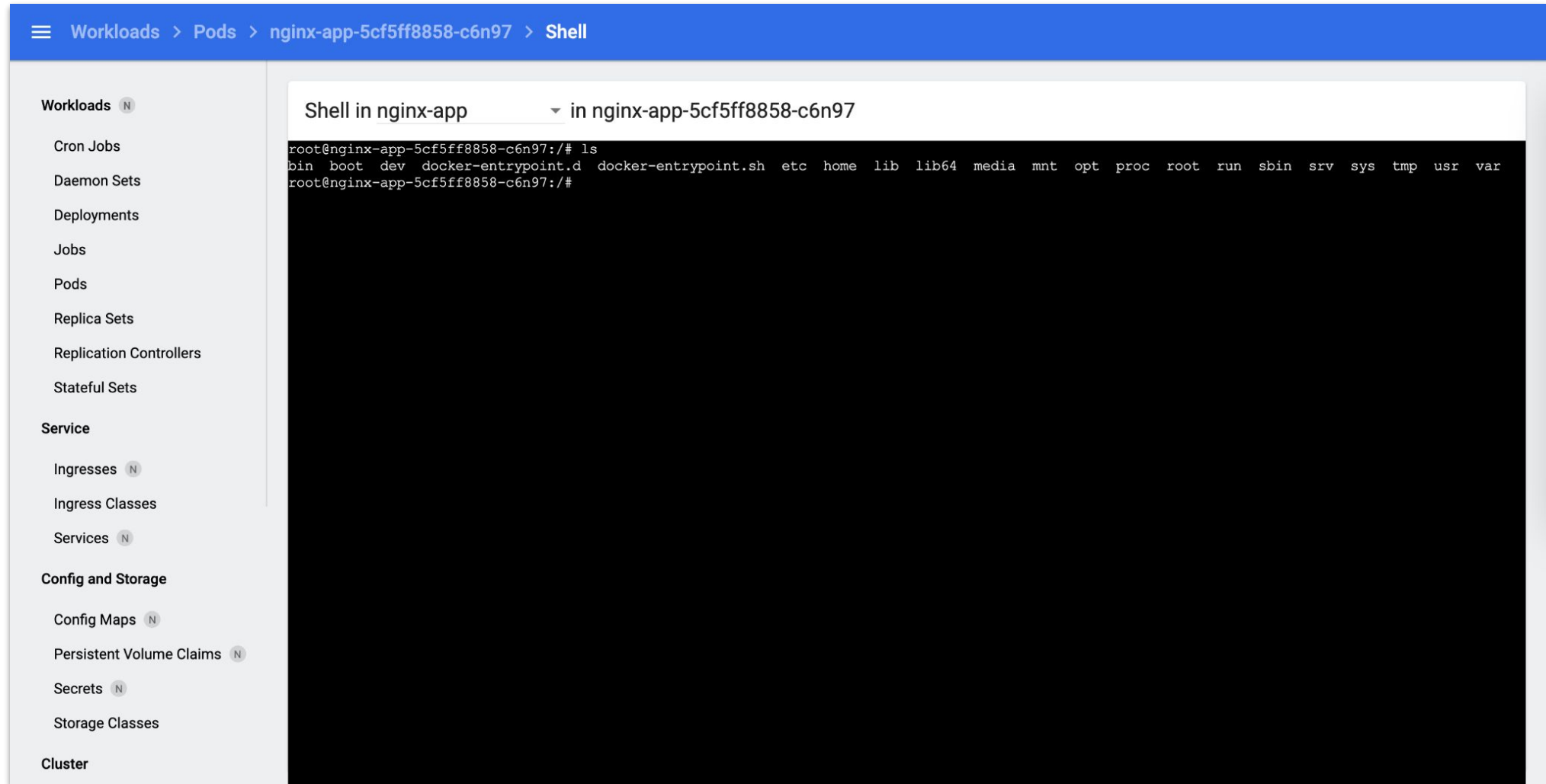
Events

Logs from nginx-app in nginx-app-5cf5...

```
2023/07/05 16:25:39 [notice] 1#1: Built by gcc 10.2.1 20210110 (Debian 10.2.1-6)
2023/07/05 16:25:39 [notice] 1#1: OS: Linux 5.19.0-1027-gcp
2023/07/05 16:25:39 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 65535:65535
2023/07/05 16:25:39 [notice] 1#1: start worker processes
2023/07/05 16:25:39 [notice] 1#1: start worker process 30
2023/07/05 16:25:39 [notice] 1#1: start worker process 31
10.233.102.128 - - [05/Jul/2023:16:33:08 +0000] "GET / HTTP/1.1" 200 615 "-" "curl/7.81.0" "-"
2023/07/05 16:58:57 [error] 31#31: *2 open() "/usr/share/nginx/html/.well-known/acme-challenge/aIP3Xj1TBfHBERbnGNuxtHs5ZPNyktGS9ieELg9iYRk" failed (2: No such crops.co", referer: "http://nginx.e-crops.co/.well-known/acme-challenge/aIP3Xj1TBfHBERbnGNuxtHs5ZPNyktGS9ieELg9iYRk"
10.233.102.139 - - [05/Jul/2023:16:58:57 +0000] "GET /.well-known/acme-challenge/aIP3Xj1TBfHBERbnGNuxtHs5ZPNyktGS9ieELg9iYRk HTTP/1.1" 404 153 "http://nginx.e-manager/e3a2a803e8ed7f8a88d5f535d6e9a061c1571194" "34.124.255.127"
10.233.102.139 - - [05/Jul/2023:17:00:32 +0000] "GET / HTTP/1.1" 200 615 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like G
10.233.102.139 - - [05/Jul/2023:17:00:38 +0000] "GET / HTTP/1.1" 200 615 "-" "Mozilla/5.0 (Windows NT 6.1; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) C
2023/07/05 17:00:40 [error] 30#30: *4 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 10.233.102.139, server: localho
10.233.102.139 - - [05/Jul/2023:17:00:40 +0000] "GET /favicon.ico HTTP/1.1" 404 153 "-" "python-requests/2.25.1" "51.81.245.138"
10.233.102.139 - - [05/Jul/2023:17:01:00 +0000] "GET / HTTP/1.1" 200 615 "-" "-" "139.144.150.26"
10.233.102.139 - - [05/Jul/2023:17:01:00 +0000] "GET / HTTP/1.1" 200 615 "-" "Mozilla/5.0 zgrab/0.x" "171.67.70.229"
10.233.102.139 - - [05/Jul/2023:17:01:05 +0000] "GET / HTTP/1.1" 200 615 "-" "Mozilla/5.0 (Linux; Android 6.0; HTC One M9 Build/MRA15474) AppleWebKit/537.36 (K
2023/07/05 17:01:07 [error] 30#30: *4 open() "/usr/share/nginx/html/about" failed (2: No such file or directory), client: 10.233.102.139, server: localhost, re
10.233.102.139 - - [05/Jul/2023:17:01:07 +0000] "GET /about HTTP/1.1" 404 153 "-" "Go-http-client/1.1" "139.144.150.26"
10.233.102.139 - - [05/Jul/2023:17:01:08 +0000] "GET / HTTP/1.1" 200 615 "-" "Mozilla/5.0 (Windows NT 6.4; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome
10.233.102.139 - - [05/Jul/2023:17:01:09 +0000] "GET /ecp/Current/exporttool/microsoft.exchange.ediscovery.exporttool.application HTTP/1.1" 404 153 "-" "Go-htt
2023/07/05 17:01:09 [error] 30#30: *4 open() "/usr/share/nginx/html/ecp/Current/exporttool/microsoft.exchange.ediscovery.exporttool.application" failed (2: No crops.co"
2023/07/05 17:01:10 [error] 30#30: *3 open() "/usr/share/nginx/html/login.action" failed (2: No such file or directory), client: 10.233.102.139, server: localh
10.233.102.139 - - [05/Jul/2023:17:01:10 +0000] "GET /login.action HTTP/1.1" 404 153 "-" "Go-http-client/1.1" "139.144.150.26"
2023/07/05 17:01:13 [error] 30#30: *4 open() "/usr/share/nginx/html/.env" failed (2: No such file or directory), client: 10.233.102.139, server: localhost, req
10.233.102.139 - - [05/Jul/2023:17:01:13 +0000] "GET /.env HTTP/1.1" 404 153 "-" "Go-http-client/1.1" "139.144.150.26"
2023/07/05 17:01:13 [error] 30#30: *3 open() "/usr/share/nginx/html/.git/config" failed (2: No such file or directory), client: 10.233.102.139, server: localho
10.233.102.139 - - [05/Jul/2023:17:01:13 +0000] "GET /.git/config HTTP/1.1" 404 153 "-" "Go-http-client/1.1" "139.144.150.26"
2023/07/05 17:01:15 [error] 30#30: *4 open() "/usr/share/nginx/html/config.json" failed (2: No such file or directory), client: 10.233.102.139, server: localho
10.233.102.139 - - [05/Jul/2023:17:01:15 +0000] "GET /config.json HTTP/1.1" 404 153 "-" "Go-http-client/1.1" "139.144.150.26"
10.233.102.139 - - [05/Jul/2023:17:01:30 +0000] "GET / HTTP/1.1" 200 615 "https://nginx.e-crops.co/" "Mozilla/5.0 (Linux; U; Android 4.4.2; en-US; HM NOTE 1W B
10.233.102.139 - - [05/Jul/2023:17:01:30 +0000] "GET / HTTP/1.1" 200 615 "https://nginx.e-crops.co/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebK
10.233.102.139 - - [05/Jul/2023:17:01:31 +0000] "GET / HTTP/1.1" 200 615 "https://nginx.e-crops.co/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537
10.233.102.139 - - [05/Jul/2023:17:01:39 +0000] "HEAD / HTTP/1.1" 200 0 "http://nginx.e-crops.co" "Go-http-client/2.0" "195.74.76.198"
10.233.102.139 - - [05/Jul/2023:17:01:39 +0000] "HEAD / HTTP/1.1" 200 0 "http://nginx.e-crops.co" "Go-http-client/2.0" "195.74.76.198"
10.233.102.139 - - [05/Jul/2023:17:24:12 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gec
```

Logs from Jul 5, 2023 to Jul 6, 2023 UTC

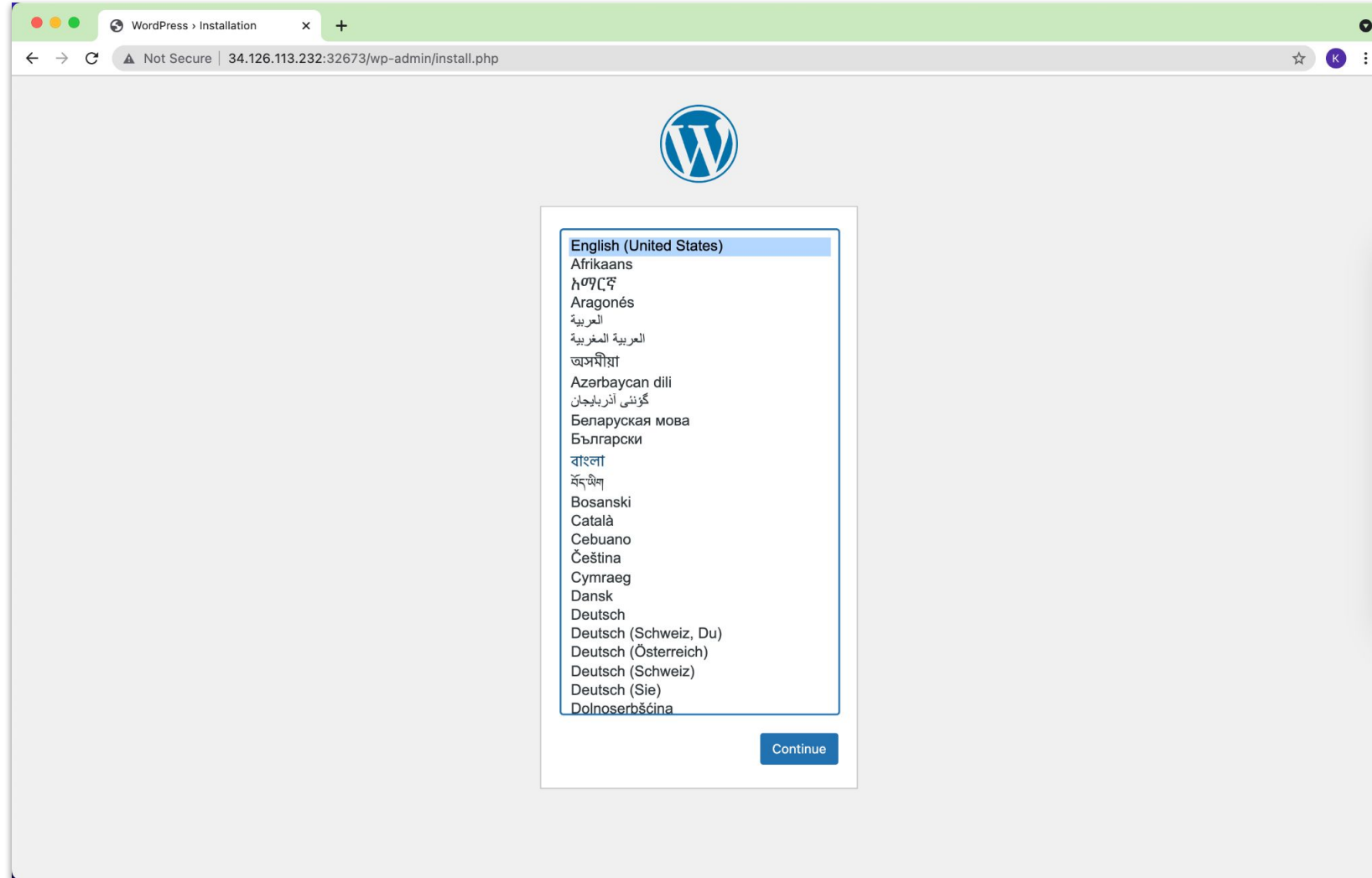
Execute into Pod



The screenshot displays the Kubernetes dashboard interface. The top navigation bar shows the breadcrumb: **Workloads > Pods > nginx-app-5cf5ff8858-c6n97 > Shell**. On the left sidebar, the 'Workloads' section is expanded, showing a list of resources: Cron Jobs, Daemon Sets, Deployments, Jobs, Pods, Replica Sets, Replication Controllers, and Stateful Sets. Below this, the 'Service' section includes Ingresses, Ingress Classes, and Services. The 'Config and Storage' section includes Config Maps, Persistent Volume Claims, Secrets, and Storage Classes. The 'Cluster' section is at the bottom. The main panel on the right is titled 'Shell in nginx-app' and shows a terminal window with the command prompt 'root@nginx-app-5cf5ff8858-c6n97:/#'. The terminal output shows the result of the 'ls' command: 'bin boot dev docker-entrypoint.d docker-entrypoint.sh etc home lib lib64 media mnt opt proc root run sbin srv sys tmp usr var'.

Practice

- Create MySQL and Wordpress for both Deployments and Services via Kubernetes Dashboard
- Resource: <https://github.com/leangsengk90/ingress-exercise.git>



A large, faint watermark of the ISTAD logo is centered in the background. It is a circular emblem with a blue outer ring containing the text 'INSTITUTE OF SCIENCE AND TECHNOLOGY ADVANCED DEVELOPMENT' and Khmer text. Inside the ring is a blue circle with a white atomic symbol. The word 'ISTAD' is written in red across the bottom of the inner circle. A blue speech bubble with three white dots is positioned over the center of the logo.

Thank you

Begin with the theory, Start with the practical